

Sheet 5 — Inner product and geometry

Numbering continues from Sheet 4. Unless stated otherwise, $\mathbb{R}^2$ and $\mathbb{R}^3$ carry the standard inner product.

- 83.** If $\mathbf{u} = (1, -2, 4)$ and $\mathbf{v} = (3, 5, 0)$, compute $\mathbf{u} \cdot \mathbf{v}$.
- 84.** Given the vectors $\mathbf{u} = (a, 2, 1)$, $\mathbf{v} = (3, 2, a)$ and the points $A(0, 1, 3)$ and $B(-1, 2, 0)$, find the value of a such that $(\mathbf{u} + \overrightarrow{AB}) \cdot \mathbf{v} = 1$.
- 85.** Given that the distance between the points $A(1, -2, 3)$ and $B(n, 2, 2)$ is $\sqrt{17}$, compute n .
- 86.** Find n such that the vector $\mathbf{u} = \left(n, 1, -\frac{1}{2}\right)$ is a unit vector.
- 87.** Consider the Euclidean space $\mathbb{R}^3$ (standard inner product).
- Show that the vectors $(2, -3, 1)$ and $(5, 5, 5)$ are orthogonal.
 - Show that, in the standard basis of $\mathbb{R}^3$, each vector is orthogonal to every linear combination of the other two.
 - Show that the vectors $(0, z, -y)$, $(z, 0, -x)$ and $(y, -x, 0)$ are orthogonal to (x, y, z) .
- 88.** Find expressions for the cosine of the angle between two vectors $\mathbf{u} = (x, y, z)$ and $\mathbf{v} = (x_1, y_1, z_1)$ and for their distance. Compute the distance between the vectors $\mathbf{a} = (3, 1, 2)$ and $\mathbf{b} = (-1, -2, -1)$ and the cosine of the angle between them.
- 89.** Show that the standard basis of $\mathbb{R}^3$ is orthonormal.
- 90.** In $\mathbb{R}^3$, with the standard inner product $(x_1, y_1, z_1) \cdot (x_2, y_2, z_2) = x_1x_2 + y_1y_2 + z_1z_2$, let $\mathbf{u} = (2, 1, 3)$, $\mathbf{v} = (-1, 4, 1)$ and $\mathbf{w} = (4, -1, -2)$. Find:
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w})$;
 - $\|2\mathbf{v} - 3\mathbf{w}\|$;
 - the cosine of the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$;
 - the distance from $\mathbf{u} + \mathbf{v}$ to $\mathbf{v} + 3\mathbf{u}$.
- 91.** In a Euclidean space U , show that if the vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ satisfy $\mathbf{u} \perp \mathbf{v}$ and $\mathbf{u} + \mathbf{v} = \mathbf{w}$, then $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{w}\|^2$.
- 92.** In the Euclidean space $\mathbb{R}^3$ (standard inner product), consider the vectors $\mathbf{u} = (1, 1, 0)$ and $\mathbf{v} = (1, -1, 0)$.
- Find the relation between x , y and z such that the vector $\mathbf{w} = (x, y, z)$ is equidistant from $\mathbf{u}$ and $\mathbf{v}$.
 - Find the unit vectors that make angles of 60° with $\mathbf{u}$ and with $\mathbf{v}$.
 - Show that the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal and find the vectors $\mathbf{t}$ orthogonal to both $\mathbf{u}$ and $\mathbf{v}$. From $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{t}$, obtain an orthonormal basis of $\mathbb{R}^3$.
- 93.** Let U be a Euclidean space with basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.
- Find the scalar α such that $\mathbf{u}_2 - \alpha\mathbf{u}_1$ is orthogonal to $\mathbf{u}_1$.
 - From $\{\mathbf{u}_1, \mathbf{u}_2\}$, obtain an orthonormal basis.
- 94.** Consider the Euclidean space $\mathbb{R}^2$ (standard inner product). Use the previous exercise to obtain an orthonormal basis from the basis $\{(1, 2), (2, 3)\}$.
- 95.** Let U be a Euclidean space with basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. Find the scalars α , β and γ such that the vectors

$$\mathbf{v}_1 = \mathbf{u}_1, \quad \mathbf{v}_2 = \mathbf{u}_2 - \alpha\mathbf{v}_1, \quad \mathbf{v}_3 = \mathbf{u}_3 - \beta\mathbf{v}_2 - \gamma\mathbf{v}_1$$

are pairwise orthogonal.

- 96.** In the Euclidean space $\mathbb{R}^3$ (standard inner product), let $B = \{(1, 1, 0), (1, 0, 1), (0, 2, 0)\}$ be a basis. Use the previous exercise to obtain, from B , an orthonormal basis of $\mathbb{R}^3$.

- 97.** Find the interior angles of the triangle ABC , where $A(3, -3, 3)$, $B(2, -1, 2)$ and $C(1, 0, 2)$.
- 98.** Given that the vector $\mathbf{v} = (2, 1, -1)$ makes an angle of 60° with the vector $\overrightarrow{AB}$ determined by the points $A(3, 1, -2)$ and $B(4, 0, m)$, compute m .

Geometry in $\mathbb{R}^3$ (new exercises, deck II: cross product, lines and planes)

- S1.** Let $\mathbf{u} = (1, -1, 2)$ and $\mathbf{v} = (3, 0, 1)$.
- Compute $\mathbf{u} \times \mathbf{v}$ and check that it is orthogonal to $\mathbf{u}$ and to $\mathbf{v}$.
 - Compute $\mathbf{v} \times \mathbf{u}$ and compare it with the previous part.
 - Compute the area of the parallelogram defined by $\mathbf{u}$ and $\mathbf{v}$.
- S2.** Compute the area of the triangle with vertices $A(1, 0, 1)$, $B(2, 1, 3)$ and $C(0, 2, 1)$.
- S3.** Show that, for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$:
- $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$;
 - $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$.
- S4.** Let $\mathbf{u} = (1, 2, 0)$, $\mathbf{v} = (0, 1, 1)$ and $\mathbf{w} = (2, 0, 3)$.
- Compute the volume of the parallelepiped defined by $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$.
 - Find k such that the vectors $(1, 2, 0)$, $(0, 1, 1)$ and $(2, 0, k)$ are coplanar.
- S5.** Consider the line r through the points $A(2, -1, 3)$ and $B(4, 0, 1)$.
- Write the vector, parametric and Cartesian equations of r .
 - Check whether the points $C(0, -2, 5)$ and $D(6, 1, 0)$ lie on r .
- S6.** Find the equation of the plane through $P(1, 2, -1)$ with normal vector $\mathbf{n} = (2, -1, 3)$.
- S7.** Find the equation of the plane through the points $A(1, 1, 0)$, $B(2, 0, 1)$ and $C(0, 1, 2)$.
- S8.** Classify the relative position of the planes, using the rank:
- $x + 2y - z = 1$ and $2x + 4y - 2z = 3$;
 - $x + y + z = 2$ and $x - y + 2z = 1$ (if they intersect in a line, write its parametric equations);
 - $x + y + z = 2$, $x - y + 2z = 1$ and $2x + y - z = 0$.
- S9.** Consider the line $r : (x, y, z) = (1, 0, 1) + t(1, 2, -1)$, $t \in \mathbb{R}$. Classify the relative position of r and each of the planes and, where applicable, find the intersection: a) $x - y + z = 5$; b) $x + y + 3z = 4$; c) $x + y + 3z = 1$.
- S10.** Consider the point $P(1, -2, 3)$ and the plane $\pi : 2x - y + 2z = 1$.
- Compute the distance from P to π .
 - Find the point of π closest to P . Hint: follow the line through P in the direction of $\mathbf{n}$.